

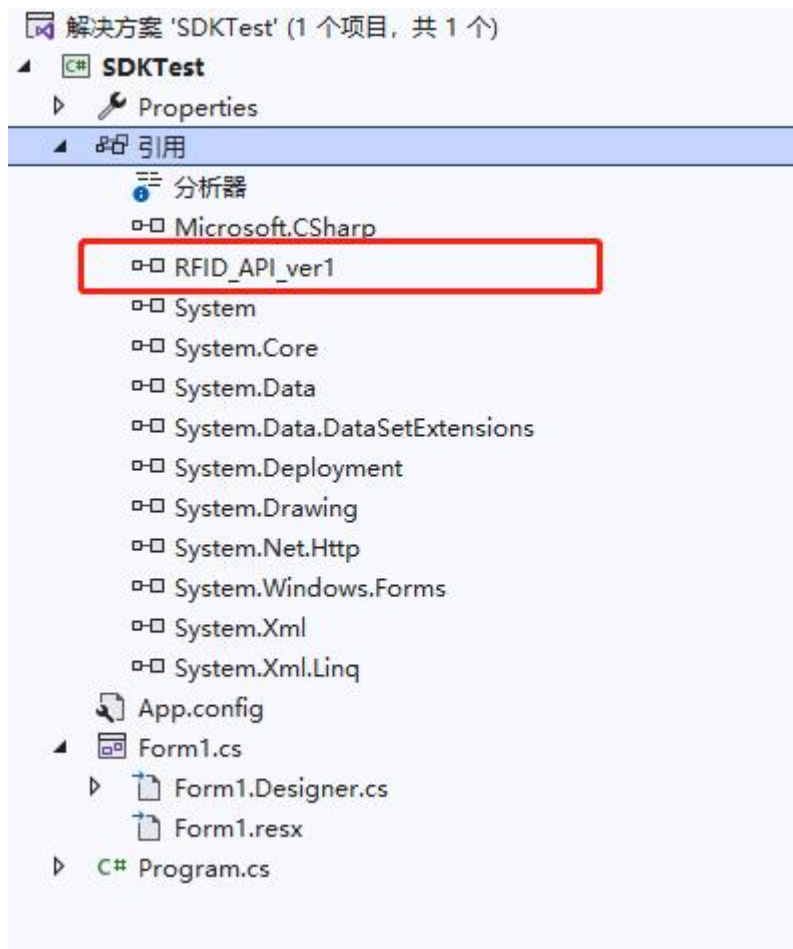
Development guide

1) List:

1. RFID_API_ver1.dll module communication control interface;
2. The C # API documentation description;
3. Development Guide;
4. Example code that the SDKTest Demo can run;

2) The C # Development Guide:

1. Add the RFID_API_ver1.dll file to the project directory by adding references, as shown in the figure:



2. After successfully referencing the RFID_API_ver1.dll file, you can declare, initialize, and invoke the Reader class that contains all the functional operation methods and data return methods for the RFID module;

(1) Instantiate the Reader class through the Create() method:

1. String mouth connection

The sample code is as follows:

```
private Reader reader ;
private ReaderType readerType = ReaderType . SERIAL; // connection type
private Channels channels = Channels . One; // Reader writer type
string arg1 = "COM2"; // string slogan
int arg2 = 115,200; // Porter rate
reader = Reader . Create (readerType, channels, arg1, arg2); // instantiate
Reader successfully instantiates reader, fails to delegate reader.
ExceptionReceived += Reader_ExceptionReceived; return the result;
```

```
private void Reader_ExceptionReceived (object sender,
ExceptionReceivedEventArgs e) // is used to return related exceptions to
the interface
{
// To achieve the corresponding operation in this method, it is necessary
to pay attention to that if you want to operate the control, it is best to
entrust the control through the operation, the error details can be viewed
through the ExceptionReceivedEventArgs class;
}
```

2. Network port connection

The sample code is as follows:

```
private Reader reader ;
private ReaderType readerType = ReaderType . TCP; // Connection Type
private Channels channels = Channels . One; // Reader writer type
string arg1 = "192.168.1.100"; // IP address (here indicates the IP address
of the device required to be connected)
int arg2 = 4001; // Port number (generally the default is 4001)
reader = Reader . Create (readerType, channels, arg1, arg2); // instantiate
Reader successfully instantiates reader, fails to delegate reader.
ExceptionReceived += Reader_ExceptionReceived; return the result;
private void Reader_ExceptionReceived (object sender,
ExceptionReceivedEventArgs e) // is used to return related exceptions to
the interface
{
// To achieve the corresponding operation in this method, it is necessary
to pay attention to that if you want to operate the control, it is best to
entrust the control through the operation, the error details can be viewed
through the ExceptionReceivedEventArgs class;
}
```

(2) After you have instantiated reader, you call reader.Connect();

(3) The relevant data return of the module function operation is returned to the upper layer application through the form of entrustment, without realizing all the entrustment methods. In addition, the operation control needs to refresh the data in the entrustment method, and it is best to operate the control in the form of entrustment;

(4) Description of various entrustment methods:

The sample code is as follows:

```
reader .TagRead += Reader_TagRead ;
reader .ReadTagCountCallback += Reader_ReadTagCountCallback ;
reader .CommandStatistics += Reader_CommandStatistics ;
reader .MessageTransport += PrintTransportLog ;
reader .ExceptionReceived += Reader_ExceptionReceived ;
reader .FirmwareVersionCallback += ReshFirmwareVersion ;
reader .FrequencyCallback += Reader_FrequencyCallback ;
```

```

reader .SettingStatusCallback += Reader_SettingStatusCallback ;
reader .AntPowerCallback += Reader_AntPowerCallback ;
reader .TagOpResultCallback += Reader_TagOpResultCallback ;
reader .ReaderIdentifierCallback += Reader_ReaderIdentifierCallback ;
reader .ReaderTemperatureCallback += Reader_ReaderTemperatureCallback ;
reader .GpioPinsCallback += Reader_GpioPinsCallback ;
reader .WorkAntennaCallback += Reader_WorkAntennaCallback ;
reader .RFPortReturnLossCallback += Reader_RFPortReturnLossCallback ;
reader .RfLinkProfileCallback += Reader_RfLinkProfileCallback ;
reader .ImpinjFastTidCallback += Reader_ImpinjFastTidCallback ;
reader .E710ProfileCallback += Reader_E710ProfileCallback ;
reader .AntConnectionDetectorCallback +=
Reader_AntConnectionDetectorCallback ;
reader .SelectCallback += Reader_SelectCallback ;
reader .QConfigCallback += Reader_QConfigCallback ;
reader .EpcMatchCallback += Reader_EpcMatchCallback ;

```

1. Working antenna data return method:

The sample code is as follows:

```

private void Reader_WorkAntennaCallback (object sender ,
WorkAntennaEventArgs e )
{
BeginInvoke (new ThreadStart (delegate ()
{
// Return the control display data by delegate, e. The ant gets the current
working antenna;
cmbWorkAnt .SelectedIndex = (int )e .ant ;
})));
}

```

2. Get the Gpio1 and Gpio2 value data return method:

The sample code is as follows:

```

private void Reader_GpioPinsCallback (object sender , GpioPinsEventArgs
e )
{
BeginInvoke (new ThreadStart (delegate ()
{
foreach (GpioPin gpi in e .gpis )
{
if (gpi . ID == 1) // gpio1 level data
{
if (gpi .High )
{
// High-level display

```

```

}
else
{
// Low level display
}
}
else if (gpi . ID == 2) // level data for / gpio1
{
if (gpi .High )
{
// High-level display
}
else
{
// Low level display
}
}
}
}));
}

```

3. Function setting of the operation data return method:

The sample code is as follows:

```

private void Reader_SettingStatusCallback (object sender ,
SettingEventArgs e )
{
// The status type shows whether the setting is successful
switch (e .status )
{
case ErrorCode .COMMAND_FAIL :
strlog = "setting failed";
break ;
case ErrorCode .COMMAND_SUCCESS :
strlog = "Set up successfully";
break ;
}
}
}

```

4. Obband operation data return method:

The sample code is as follows:

```

private void Reader_FrequencyCallback (object sender , FreqEventArgs
e )
{
BeginInit (new ThreadStart (delegate ()

```

```

{
// Obtain the frequency band-related data
FrequencyRegion frequencyRegion = e .FreqRegion ;

switch (frequencyRegion .Region )
{
case RFID_API_ver1.Region .FCC :

break ;
case RFID_API_ver1.Region .ETSI :

break ;
case RFID_API_ver1.Region .CHN :

break ;
case RFID_API_ver1.Region .CUSTOMIZE :

break ;
default :
break ;
}
}));
}

```

5. Get the firmware version data return method:

The sample code is as follows:

```

private void ReshFirmwareVersion (object sender ,
FirmwareVersionEventArgs e )
{
BeginInvoke (new ThreadStart (delegate ()
{
string fwVer = string .Format ("{0:x2}.{1:x2}", e .Info .Major ,
e .Info .Minor );
txtFirmwareVersion .Text = fwVer ;
}));
}

```

6. Serial print data return method: (serial print data is time-consuming, if you want to refresh to the control, consider the display real-time problem)

The sample code is as follows:

```

private void PrintTransportLog (object sender ,
MessageTransportEventArgs e )
{

```

```

string strLog = string.Format("{0:5}: {1}", (e.Tx ? "Send" : "Recv"),
ByteUtils.ToHex(e.TransportData, "", " "));
}

```

7. Return method of obtaining antenna power data:

The sample code is as follows:

```

private void Reader_AntPowerCallback(object sender, AntPowerEventArgs
e )
{
e.power[i] // antenna power return data
}

```

8. Label operation (read and write operation) data return method:

The sample code is as follows:

```

private void Reader_TagOpResultCallback(object sender,
TagOpResultEventArgs e )
{
// Data display operation
UpdateTagResultsDgv(e.tagOps);
}

```

9. Obtain the module operating temperature data return method:

The sample code is as follows:

```

private void Reader_ReaderTemperatureCallback(object sender,
ReaderTemperatureEventArgs e )
{
BeginInvoke(new ThreadStart(delegate ()
{
// Temperature display operation
string.Format("{0} °C", e.ReaderTemperature);
})));
}

```

10. Obtain the echo loss data return method:

The sample code is as follows:

```

private void Reader_RFPortReturnLossCallback(object sender,
RfPortReturnLossEventArgs e )
{
BeginInvoke(new ThreadStart(delegate ()
{
// return loss
string.Format("{0}", e.Loss);
})));
}

```

11. Obtaining the link data return method:

The sample code is as follows:

```
private void Reader_RfLinkProfileCallback (object sender ,
RfLinkProfileEventArgs e )
{
BeginInvoke (new ThreadStart (delegate ()
{
// The Link returns the result
e .RfLinkProfile ;
})));
}
```

12. Obtain the FastID data return method:

The sample code is as follows:

```
private void Reader_ImpinjFastTidCallback (object sender ,
ImpinjFastTidEventArgs e )
{
BeginInvoke (new ThreadStart (delegate ()
{
if (e .ImpinjFastTid )
{
// open
}
else
{
// close
}
})));
}
```

13. Obtaining the E710 link data return method:

The sample code is as follows:

```
private void Reader_E710ProfileCallback (object sender ,
E710ProfileEventArgs e )
{
BeginInvoke (new ThreadStart (delegate ()
{
// E710 Link result return
})));
}
```

14. Obtaining the antenna detection and return method:

The sample code is as follows:


```

private void Reader_AntConnectionDetectorCallback (object sender ,
AntConnectionDetectorEventArgs e )
{
BeginInvoke (new ThreadStart (delegate ()
{
// Antenna test result
string .Format ("{0}", e .AntConnectionDetector );
}));
}

```

The 15.Select operation data return method:

The sample code is as follows:

```

private void Reader_SelectCallback (object sender , SelectEventArgs
e )
{
BeginInvoke (new ThreadStart (delegate ()
{
string strResults = "";
tagMaskDB .Clear ();
foreach (Select select in e .selects )
{
strResults = string .Format ("{0}\r\n {1};", strResults , select .ToString
());
tagMaskDB .Add (select );
}
}));
}

```

16. One disk result data return method: (this method receives data represents the end of one inventory, can perform the next inventory)

The sample code is as follows:

```

private void Reader_CommandStatistics (object sender ,
CommandstatisticsEventArgs e )
{
this .BeginInvoke (new ThreadStart (delegate ()
{
// Single inventory results are returned
string .Format ("{0}", e .RoundTotalRead );
string .Format ("{0}", e .RoundReadRate );
string .Format ("{0}", e .RoundCommandDuration );
}));
}

```

17. Save label data return method: (After starting begins, label data is returned by this

method, returning data for every label read)

The sample code is as follows:

```
private void Reader_TagRead (object sender , TagReadEventArgs e )
{
    BeginInvoke (new ThreadStart (delegate ()
    {
        // Tagged data
        e .TagData ;

    }));
}
```

18. Obtaining the E710Q value data return method:

The sample code is as follows:

```
private void Reader_QConfigCallback (object sender , QConfigEventArgs e )
{
    // Operation controls should preferably use a delegate method
    BeginInvoke (new ThreadStart (delegate ()
    {
        // Q value data
        if (e .config .UseAdvanceConfig )
        {
            e .config .QInit ;
            e .config .QMin ;
            e .config .QMax ;
            e .config .NumMinQCycles ;
            e .config .MaxQuerySinceEPC ;
        }
        else
        {
            e .config .QInit ;
            e .config .QMin ;
            e .config .QMax ;
        }
    }));
}
```

3. Resource release, after calling the delegate method, the corresponding delegate method:

The sample code is as follows:

```
reader .TagRead -= Reader_TagRead ;
```

```

reader .ReadTagCountCallback  -= Reader_ReadTagCountCallback ;
reader .CommandStatistics    -= Reader_CommandStatistics ;
reader .MessageTransport     -= PrintTransportLog ;
reader .ExceptionReceived    -= Reader_ExceptionReceived ;
reader .FirmwareVersionCallback  -= ReshFirmwareVersion ;
reader .FrequencyCallback     -= Reader_FrequencyCallback ;
reader .SettingStatusCallback  -= Reader_SettingStatusCallback ;
reader .AntPowerCallback      -= Reader_AntPowerCallback ;
reader .TagOpResultCallback    -= Reader_TagOpResultCallback ;
reader .ReaderIdentifierCallback -= Reader_ReaderIdentifierCallback ;
reader .ReaderTemperatureCallback -= Reader_ReaderTemperatureCallback ;
reader .GpioPinsCallback      -= Reader_GpioPinsCallback ;
reader .WorkAntennaCallback    -= Reader_WorkAntennaCallback ;
reader .RFPortReturnLossCallback -= Reader_RFPortReturnLossCallback ;
reader .RfLinkProfileCallback  -= Reader_RfLinkProfileCallback ;
reader .ImpinjFastTidCallback  -= Reader_ImpinjFastTidCallback ;
reader .E710ProfileCallback    -= Reader_E710ProfileCallback ;
reader .AntConnectionDetectorCallback -=
Reader_AntConnectionDetectorCallback ;
reader .SelectCallback        -= Reader_SelectCallback ;
reader .QConfigCallback       -= Reader_QConfigCallback ;
reader .EpcMatchCallback      -= Reader_EpcMatchCallback ;

```

Finally by calling the reader. Disconnect (); method to release the serial port resources;